

ALEX SALAMATI

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EDUCATION

Loyola Marymount University, Los Angeles, CA |

Expected May 2027

B.S. Computer Engineering, Minor: Computer Science

Relevant Coursework: Data Structures & Algorithms, Computer Architecture (SystemVerilog), Embedded Systems, Microprocessors/Microcontrollers, Circuits I & II, Electronics, Digital Systems, Digital & Electronic Design I & II

TECHNICAL SKILLS

Languages: Python, C, C++, MATLAB, SystemVerilog, Verilog, Bash

Hardware & Protocols: FPGA (Xilinx), Arduino, Embedded Systems, I2C, SPI, UART, Oscilloscopes, Multimeters

Tools & Frameworks: Git/GitHub, Linux, OpenCV, MediaPipe, ROS (learning), Vivado, Multisim, CAD/3D Printing

Concepts: PID Control, Computer Vision, RTL Design, Digital Signal Processing, Circuit Analysis, DSA

PROJECTS

Ball Balancing Robot *PID Control | Computer Vision | Python | OpenCV | Servo Systems*

Mar 2026 – Present

- Designing a closed-loop 3-DOF servo-driven platform that autonomously balances a ball by dynamically tilting the surface in real time
- Built an OpenCV + MediaPipe vision pipeline for ball detection and velocity estimation at 30+ FPS, feeding position data into a PID controller
- Tuning PID gains (Kp, Ki, Kd) using step-response analysis to minimize overshoot and achieve stable, low-latency ball tracking
- Integrating hardware (servo PWM control via Arduino) with software (Python control loop) into a full embedded feedback system

MIPS Single-Cycle Processor *RTL Design | SystemVerilog | FPGA | Xilinx Vivado*

Mar – May 2026

- Designed a complete single-cycle MIPS processor in SystemVerilog, including ALU, control unit, register file, instruction memory, and full datapath
- Synthesized and deployed on Xilinx FPGA using Vivado; performed RTL simulation, timing analysis, and verified setup/hold constraints across 15+ instructions
- Wrote comprehensive testbenches covering R-type, I-type, and J-type instructions; confirmed correct branch, jump, and memory operation behavior

Custom Two-Stage CMOS Op-Amp *Analog IC Design | Frequency Compensation | Multisim*

Feb – Mar 2026

- Designed a two-stage CMOS op-amp from the transistor level to meet high open-loop gain (>60dB) and low input offset voltage specifications
- Applied Miller compensation capacitor to achieve >45° phase margin; verified closed-loop stability via Bode plots and AC/DC simulation in Multisim

Real-Time Hand Gesture Tracker *Computer Vision | MediaPipe | Python | OpenCV*

May – Jul 2025

- Built a real-time hand landmark detection system using MediaPipe and OpenCV; classified raised fingers using landmark coordinate geometry
- Displayed live finger-count overlay on video feed at 30+ FPS; foundation for gesture-based HCI and future robotics control interface

EXPERIENCE

IT Support Technician *LMU ITS Classroom Support | Work Study*

Jun 2024 – Present

- Maintain and troubleshoot AV, OS, and hardware systems across 20+ smart classrooms; diagnose and resolve failures in under 5 minutes during live lectures
- Debug cross-platform issues across Windows and macOS; support faculty with projectors, microphones, lecture capture, and network connectivity
- Document recurring hardware failures and escalate firmware-level issues to vendors, reducing repeat incidents by improving issue tracking workflow

LEADERSHIP & ACTIVITIES

Vice President, *Additive Manufacturing Club, LMU*

2024 – Present

- Lead end-to-end 3D printing projects: manage CAD design reviews, slicing parameter optimization, and printer calibration for FDM and resin systems
- Train new members on design-for-manufacturing principles, CAD validation workflows, and material selection; grew active membership by onboarding 10+ new students

Member, *Robotics Optimization & Building Organization (ROBO), LMU*

2023 – Present

- Collaborate on cross-disciplinary robotics builds, integrating sensors, motors, embedded control systems, and mechanical design
- Contributed to an autonomous navigation project exploring obstacle avoidance using ultrasonic sensors and microcontroller-based decision logic

Member, *IEEE Student Chapter, LMU*

2023 – Present

- Attend technical workshops on embedded systems, PCB design, and signal processing; engage with industry professionals at networking events